

Can AI agents measure accessibility?

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Abstract:

Thousands of specialized tools can detect and diagnose defects in software quality. Large language models are no substitute for those tools. Quality testing by LLMs alone produces speculative, cursory, fragmentary, vague, and/or fabricated results. To be effective, LLMs need tools. The LLM job is to understand user requests, identify appropriate tools, formulate inputs to the tools, evaluate tool outputs, and convert them into appropriate responses to users.

Web accessibility illustrates this problem. When asked to assess the accessibility of a web page, an LLM struggles to (1) recognize its own need for tools, (2) discover tools, (3) choose appropriate tools, and (4) become authorized to use them. Users may need to help an LLM accomplish all four of these steps. Success can dramatically improve results.

Thus, AI can achieve quality in the assessment of software quality, but only if human-directed, not autonomous.

I. Problem

AI autonomy complicates the assessment of software quality. There is not only a test suite to perfect, but also an AI agent to trust or distrust. A large language model may try to audit on its own, or may look for appropriate tools among the thousands that can detect and diagnose defects. It makes good or bad decisions: whether to call tools at all, how to find tools, which ones to call, what requests to submit to them, how to evaluate their outputs, and how to convert them into user-friendly responses. And it may be unable to execute even good decisions because of platform constraints.

Web accessibility illustrates the problem. LLMs, asked to assess the accessibility of a web page, struggle with these decisions. They often try to perform an assessment tool-free and produce cursory, fragmentary, vague, wrong, and/or fabricated responses.^{1,2} If they decide (or are instructed) to use tools, LLMs often fail to find any that they can use or choose among them with no disclosed criteria. And LLMs may be unable to call tools without user help.

II. Solutions

In these examples, a user asks Claude Sonnet 5 via Perplexity to report accessibility defects on the home page of PNSQC.

A. Total autonomy

User-defined task

“Are there accessibility defects on the home page of the Pacific Northwest Software Quality Conference? If so, what are they?”

LLM response

Called no tool, but described 4 “**likely**” issues:

- Non-descriptive link text
- Unlabeled or ambiguous ‘RSVP’ links
- Dynamic countdown timer without text alternative
- Repetitive heading/label pairs

B. Suggested tool discovery

User-defined task

“Are there accessibility defects on the home page of the Pacific Northwest Software Quality Conference? If so, what are they? If you can find any tools³ that will make your list more complete and accurate, use them.”

LLM response

Tried and failed to call 3 tools, but described 6 “**likely**” issues:

- Image alt text
- Color contrast ratios for text, buttons, and the countdown timer
- Keyboard navigation and focus indicators for interactive elements
- ARIA labeling on the live countdown timer
- Form label association for the newsletter signup field
- HTML lang attribute and semantic landmark regions

C. Directed tool discovery and enablement

User-defined task

“Are there accessibility defects on the home page of the Pacific Northwest Software Quality Conference? If so, what are they? The Kilotest⁴ MCP server, connected to your client, should be able to help make your list more complete and accurate.”

LLM response

Successfully called Kilotest tools and described 30 **attested** issues with 4 priorities, “highest”, “high”, “low”, and “lowest”.

Interpretive summary: “the highest- and high-priority findings (unnamed iframe, invalid parent structure, unnamed SVG, autocomplete errors, keyboard-operability conflicts, and target-size/spacing problems) are the most impactful for people using screen readers, keyboard navigation, or those with motor or vision impairments, and should be prioritized”

<u>Solution</u>	<u>Findings</u>	
Autonomous	4	Speculative issues
Suggested	6	Speculative issues
Directed	30	Attested issues

¹Kris Rivenburgh, “Can ChatGPT Audit Your Website for Accessibility?”, 2025. <https://accessible.org/chatgpt-audit-website-accessibility/>.
²Mingyuan Zhong *et al.*, “Targeted Interaction Agents for Web Accessibility Evaluation”, 2026. *UIST Adjunct ’26*. <https://faculty.washington.edu/wobbrock/pubs/uist-26.01.pdf>.
³No specific tool was named or pre-authorized.
⁴The Kilotest (<https://kilotest.com>) MCP server provides tools to LLMs for testing and reporting web accessibility, usability, and standards conformity.

III. Conclusion

Large language models need specialized tools to assess the accessibility, and more generally the quality, of software. When LLMs use the right tools, assessments are dramatically more comprehensive and reliable. But tool discovery, selection, and enablement require user help. In software quality assessment, **autonomy impairs quality**.